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Paleomagnetism and Fission-Track Geochronology on the Goto and Tsushima Islands in the Tsushima Strait Area: Implications for the Opening Mode of the Japan Sea

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Paleomagnetism of Neogene rocks in the Goto Islands and fission-track (FT) geochronology of Miocene igneous rocks in the Goto and Tsushima Islands were investigated in order to reveal Miocene tectonics in the Tsushima Strait area. Untilted directions of primary magnetic components from early to middle Miocene sedimentary rocks in the Goto Islands dominantly showed counter-clockwise (CCW) deflections to the expected direction of the geocentric axial dipole field, while paleomagnetic directions from Miocene igneous rocks and Quaternary basalts were concordant with the expected direction. Zircon FT ages determined on sixteen Miocene igneous rocks show good agreement at about 15 Ma. These results, in conjunction with previously-reported paleomagnetic data in the Tsushima Islands, suggest that the Goto Islands were rotated at the early to middle Miocene before about 15 Ma and that the CCW of the Tsushima islands occurred after about 15 Ma. The CCW rotations of these islands imply that the Tsushima Strait area did not belong to the Southwest Japan block in terms of clock-wise (CW) rotation at about 15 Ma, constraining the western margin of the CW-rotated block. The FT ages confine the time of the compressive deformation of pre-middle Miocene sediments in the area to be at the early to middle Miocene before about 15 Ma. The CCW rotations of these islands probably took place in response to the movement of the fault system in the area associated with the compressive regime, especially a sinistral motion of the Tsushima-Goto tectonic line. The early to middle Miocene compressive deformation in the area occurred almost coeval with the opening of the Japan Sea. The compressive tectonic regime implies the convergence of the western margin of the Southwest Japan block to the Korean Peninsula during the CW rotation of the block at about 15 Ma, which gives a new constraint on the position of the rotation pivot of the block.

1. Introduction

The Japan Sea—Japanese island-arcs is one of back-arc basin—island-arc systems on the eastern margin of Eurasia. The ages and opening histories of back-arc basins have been ordinarily revealed by means of identifications of magnetic anomaly stripes in the basins (e.g., WEISSEL, 1981). However, the identification of magnetic anomalies in the Japan Sea is still controversial because the linearity of magnetic anomalies is poor (KONO, 1987). The opening process of the Japan Sea has been studied from the view point of the drifting mode of the Japanese island-arcs through on-land paleomagnetism (OTOFUJI and MATSUDA, 1983, 1984, 1987; HAYASHIDA and ITO, 1984; OTOFUJI *et al.*, 1985a, 1985b, 1985c; HAYASHIDA, 1986; TOSHA and HAMONO, 1988; ITOH and ITO,

1989). The paleomagnetism investigations revealed that the southwestern part of the arcs, Southwest Japan, was rotated clockwise (CW) by about 47° relative to Eurasia at about 15 Ma (OTOFUJI *et al.*, 1985c). Southwest Japan has been regarded as a single drifted coherent block because pre-Neogene geologic units in the block show ENE-WSW trending zonal arrangement (SHIMAZAKI *et al.*, 1981; OZAWA *et al.*, 1985; Fig. 1(a)). The position of the rotation pivot of the block was assumed at somewhere near the edge of the block, between the Korean Peninsula and Kyushu Island (OTOFUJI and MATSUDA, 1983). The CW rotation of Southwest Japan suggested the fan-shape opening of the southwestern part of the Japan Sea.

The position of the pivot (34°N, 129°E) estimated by OTOFUJI and MATSUDA (1983) implied that the Southwest Japan block was located close to the Asian continent prior to its CW rotation (OTOFUJI and MATSUDA, 1983; HAYASHIDA and TORII, 1988). The timing of the CW rotation has been regarded as the timing of the opening of the Japan Sea. However, older opening ages have been suggested on the basis of geophysical evidence from the Japan Sea; TAMAKI (1986) suggested an age range from 30 Ma, or older to 15 Ma, for the Japan and Yamato Basins based on bathymetric and heat flow data, and ISEZAKI (1986) fitted the magnetic anomalies in the Japan Basin with a model based on the polarity reversals from 19 Ma to 15 Ma. Marine sediments of Early Miocene

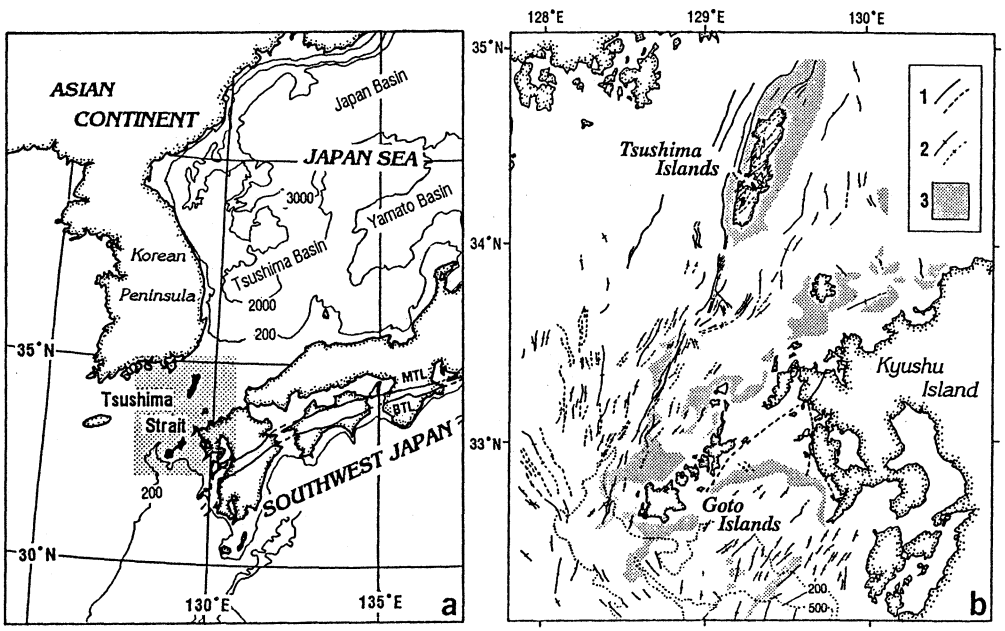


Fig. 1. (a) Map showing the present configuration of the island-arc–back-arc basin system around the western part of the Japan Sea. Base map shows bathymetric contours of 200 m, 2000 m and 3000 m, and positions of back-arc basins. Median Tectonic Line (MTL) and Butsuzo Tectonic Line (BTL) represent the zonal structure of pre-Neogene rocks in Southwest Japan. The shaded area shows the Tsushima Strait area. Black portions in the area indicate the Tsushima and Goto islands. (b) Simplified Geological structures in the Tsushima Strait area (modified from TOMITA *et al.* (1975), INOUE (1975), and NAGANO *et al.* (1976)). 1: faults. 2: fold axis. 3: pre-middle Miocene sediments exposed in the sea area which are corresponded to the sediments of the Taishu or Goto Group.

age distributed on the Japan Sea coast of the Japanese island-arcs implies the initiation of the Japan Sea opening prior to the opening accompanied with the CW rotation of the Southwest Japan block at about 15 Ma (KANO and YANAGISAWA, 1989). Hence, it is indispensable to clarify the drifting mode of Southwest Japan in more detail, especially the extent of the CW-rotated block and the preferable position of the rotation pivot, for the better understanding of the whole process of the opening of the Japan Sea.

The Tsushima Strait area between the Korean Peninsula and Kyushu Island is regarded as a boundary region between the stable Asian continent and the CW-rotated Southwest Japan block during the formation of the Japan Sea (Fig. 1). Miocene tectonics in the area must have been controlled by a relative motion between the two blocks. To reveal Miocene tectonics in the area will give a clue for clarifying the drifting mode of Southwest Japan, especially estimating the western extent of the block and the position of the rotation pivot. A previous paleomagnetic study on the Tsushima Islands in the area suggested that the islands were rotated counter-clockwise (CCW) by about 28° sometime after the middle Miocene igneous activity on the islands (ISHIKAWA *et al.*, 1989), which implies a different tectonic regime in the Tsushima Strait area from that of Southwest Japan during the formation process of the Japan Sea. In order to reveal Miocene tectonics in the Tsushima Strait area, we carried out a paleomagnetic study on Neogene rocks in the Goto Islands, another archipelago in the area, and fission-track analysis of Miocene igneous rocks in the Goto and Tsushima Islands.

2. Geological Setting and Sampling

The Goto Islands are comprised of five main islands as follows; Fukue, Hisaka, Naru, Wakamatsu, and Nakadori Islands (Fig. 2). The stratigraphic succession of Neogene rocks distributed on the islands is divided into the following four units; early to middle Miocene non-marine sediments (the Goto Group), middle to late Miocene volcanic rocks (the Miocene volcanic rocks), middle to late Miocene intrusive rocks (the Miocene intrusive rocks), and the Quaternary basalts (UEDA, 1961; KAMADA, 1966; TESHIMA and YAMAMOTO, 1972; MATSUI *et al.*, 1977; KAWAHARA *et al.*, 1984; MATSUI and KAWADA, 1986; Figs. 2 and 3).

The Goto Group consists of sandstones and mudstones with intercalated tuffaceous beds. The Goto Group has been considered as early to middle Miocene in age according to the occurrence of some plant fossils of the Daijima-type Flora and the fresh water molluscan fossils of the Nojima Fauna (UEDA, 1961; NAGAHAMA and MIZUNO, 1965; MATSUI *et al.*, 1977). The fold structure of the group shows mainly a NNE-SSW to NE-SW trend, and faults in the group are classified into NW-SE, N-S and NE-SW trending fault systems (UEDA, 1961; TESHIMA and YAMAMOTO, 1972; KAWAHARA *et al.*, 1984).

The Miocene volcanic rocks consist of tuffs, tuff brecciaes, welded tuffs, and lavas. The volcanic rocks unconformably overlie the deformed Goto Group. The lowermost part of the volcanic rocks on Nakadori and Wakamatsu Islands includes ill-sorted massive sandstone and massive mudstone beds (TESHIMA and YAMAMOTO, 1972; KAWAHARA *et al.*, 1984; Figs. 2 and 3). Marine molluscan assemblage of middle to late Miocene in age was found in the mudstone bed on Nakadori Island (KAWAHARA *et al.*, 1984).

Various types of the Miocene intrusive rocks (granite, quartz porphyry, diorite,

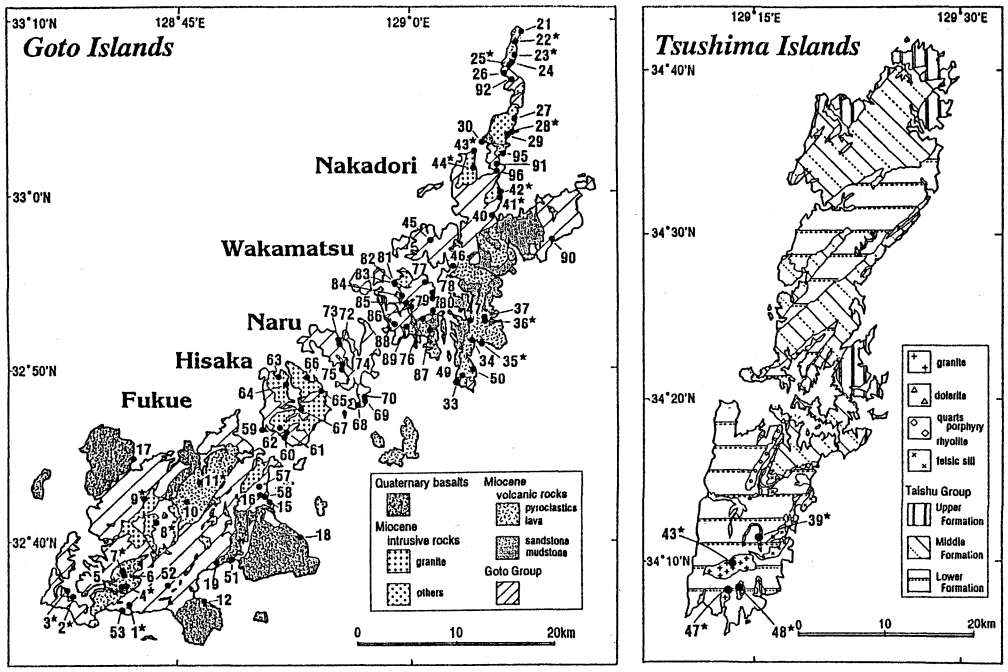


Fig. 2. Geologic maps of the Goto Islands (simplified from TESHIMA and YAMAMOTO (1972), KAWAHARA *et al.* (1984) and MATSUI and KAWADA (1986)), and the Tsushima Islands (simplified from SHIMADA (1977) and KOGA (1982)). Solid circles with numerals indicate locations of paleomagnetic sampling sites, and the numerals with star symbols indicate the sites where fission-track dating studies were also performed.

dolerite, andesite and rhyolite) intrude into the above-mentioned Neogene system in the form of dikes and stocks. Igneous activity of the intrusive rocks occurred during and after that of the Miocene volcanic rocks (UEDA, 1961; TESHIMA and YAMAMOTO, 1972; KAWAHARA *et al.*, 1984). Intrusion of granite stocks occurred at the latest stage of the igneous activity of the Miocene intrusive rocks (KAWAHARA *et al.*, 1984; MATSUI and KAWADA, 1986).

The Quaternary basalts are distributed as lava flows on Fukue island and at the northern part of Nakadori island (Fig. 2).

On the Goto Islands, we collected samples for paleomagnetic measurement at 82 sites (Fig. 2); 26 sites from sedimentary rocks of the Goto Group, 15 sites from welded tuffs of the Miocene volcanic rocks, 36 sites from the Miocene intrusive rocks (19 sites from granite stocks and 17 sites from other intrusive rocks), and 5 sites from Quaternary basalts. Six to twelve hand samples were oriented using a magnetic compass at each site. Strike and dip angles of bedding planes were measured for tilt correction whenever possible. The bedding plane of welded tuff was recognized from eutaxitic structure of the granules of pumice. We collected samples from dikes when the contact planes were approximately vertical. Fission-track (FT) dating study was also performed on 18 sites among the 82 sites (Fig. 2); 6 sites from welded tuffs of the Miocene volcanic rocks and 12 sites from the Miocene intrusive rocks (8 sites from granites and 4 sites from other intrusive rocks).

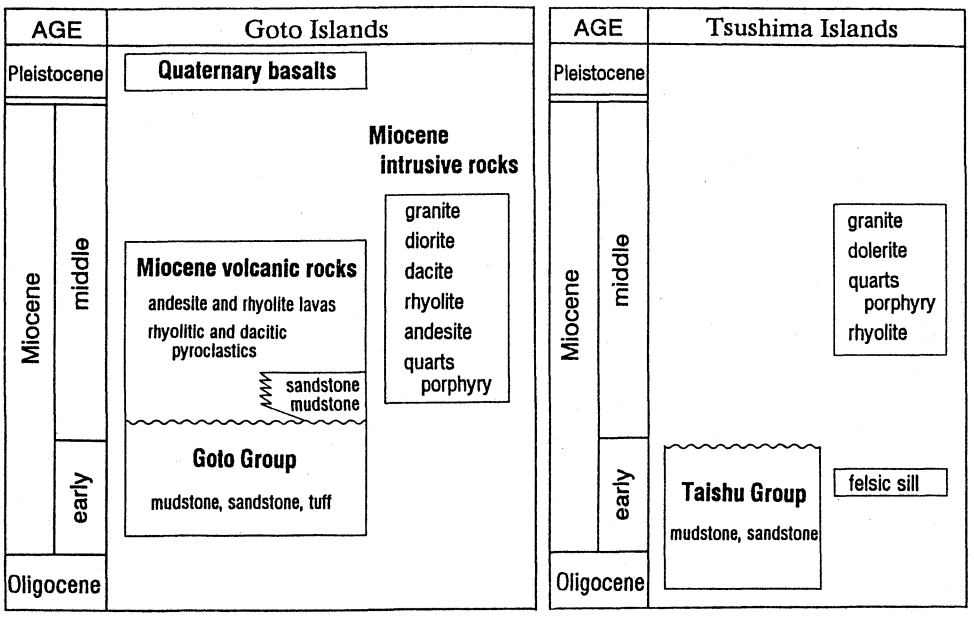


Fig. 3. Diagrams showing stratigraphic relations of the rock units in the Goto and the Tsushima Islands, respectively (complied from UEDA (1961), NAGAHAMA and MIZUNO (1965), KAMADA (1966), KAWANO and UEDA (1966), TESHIMA and YAMAMOTO (1972), MATSUI *et al.* (1977), SHIMADA (1977), KAWAHARA *et al.* (1984), TAKAHASHI and HAYASHI (1985, 1987), and MATSUI and KAWADA (1986)).

On the Tsushima Islands, thick marine sediments of Early Oligocene or older to early Miocene in age, namely the Taishu Group, and middle Miocene igneous rocks are distributed (SHIMADA, 1977; SAKAI and NISHI, 1990; Figs. 2 and 3). The Taishu Group is intensively deformed by folding and faulting. The fold axes generally trend NE-SW, disposing left-hand echeron. The faults truncating the fold system are classified into the following three types; N-S to NNE-SSW trending left-lateral strike-slip faults, NE-SW trending reverse faults, and NW-SE trending faults with vertical dips (SHIMADA, 1977). Among the middle Miocene igneous rocks, felsic sills are folded concordantly with the Taishu Group. Other intrusive rocks intersect the fold system of the group (SHIMADA, 1977). Intrusion of the granite occurred at the latest stage of the post-folding igneous activity on the islands (SHIMADA, 1977). Radiometric ages using the K–Ar and FT methods have been previously reported on the post-folding igneous rocks as follows; a single K–Ar (biotite) age of 12.3 Ma with 5% error on granite by KAWANO and UEDA (1964, 1966) (recalculated by the authors due to the revision of radiometric constants (STEIGER and JÄGER, 1977)) and three zircon FT ages of 14.2 ± 1.5 Ma on the quartz porphyry, 14.8 ± 1.6 Ma on the rhyolite, and 14.9 ± 1.6 Ma on the granite by TAKAHASHI and HAYASHI (1985, 1987) (quoted errors are 2σ standard errors).

A paleomagnetic work in the Tsushima Islands has been already carried out by ISHIKAWA *et al.* (1989). They obtained CCW-deflected paleomagnetic directions from sedimentary rocks of the Taishu Group and middle Miocene igneous rocks. The paleomagnetic results suggested a CCW rotation of the islands through about 28° after the post-folding igneous activity in middle Miocene time. To confirm the simultaneity of

tectonic events in the Tsushima Straits area, we planed to make age determinations on the post-folding igneous activity on the Tsushima Islands. The FT dating study was performed on 4 sites of the granites at the southernmost part of the islands (Fig. 2), where ISHIKAWA *et al.* (1989) had obtained paleomagnetic data.

3. Experimental Results

3.1 Paleomagnetic measurements

3.1.1. Demagnetization experiments

Two cylindrical specimens of 24 mm in diameter and 22 mm in height were prepared from each hand sample for the paleomagnetic measurements. Remanent magnetization was measured on a cryogenic magnetometer (ScT C-112) and/or on a spinner magnetometer (Schonstedt SSM-1A). The stability of the natural remanent magnetization (NRM) was examined through progressive demagnetization experiments by thermal and alternating field (AF) methods. Pilot specimens of two or three independently-oriented samples from each site were subjected to progressive demagnetizations. Stable remanent magnetization components were selected on the basis of the straight linear trend of demagnetization curve on vector-demagnetization diagram (ZIJDERVELD, 1967).

Pilot specimens of all Quaternary basalt sites and some Miocene igneous rock sites yielded essentially single stable magnetic components, except for small viscous remanence (VRM) components which were easily removed by low-demagnetization treatment below 320° C and/or 20 mT (Fig. 4-1). The stable component was clearly recognized as a straight line trend decaying toward the origin on the diagrams. The stable component was obtained by both, thermal and AF treatments (Fig. 4-1A), or by thermal treatments more effectively (Fig. 4-1B).

Most of the pilot specimens, except for the above-mentioned sites, provided two types of stable components which did not decay toward the origin of the diagram during thermal demagnetization experiments. One was often isolated in a lower-temperature range; below 280° C to 320° C for sedimentary rock specimens and below 400° C or 500° C for igneous rock ones (Fig. 4-2). The lower-temperature component was characterized by a normal polarity and approximately northerly declination in in-situ coordinates. The other type was found in the intermediate temperature range approximately between 300° C and 500 or 540° C from NRM of pilot specimens from six sites of the Goto Group (sites 24, 26, 67, 77, 92 and 96; Fig. 4-3A and B) and two sites of the welded tuff (sites 4 and 5; Fig. 4-3C). The moderate-temperature components had a southerly declination with reverse polarity in in-situ coordinates.

The lower-temperature component was generally close to the direction of the present geomagnetic field. The lower-temperature component was suspected as a secondary component acquired under the recent geomagnetic field. Site-mean directions of the moderate-temperature components were determined by applying a principal component analysis (KIRSCHVINK, 1980) on progressive thermal demagnetization results from all specimens of the above-mentioned eight sites. The site-mean directions of the moderate-temperature components made a tight cluster before the tilt correction (Fig. 4-3D). The middle-temperature components were regarded as secondary components acquired after tilting.

Most of sedimentary rock specimens provided erratic behaviors of magnetization in the higher temperature range after the lower- and/or the moderate-temperature compo-

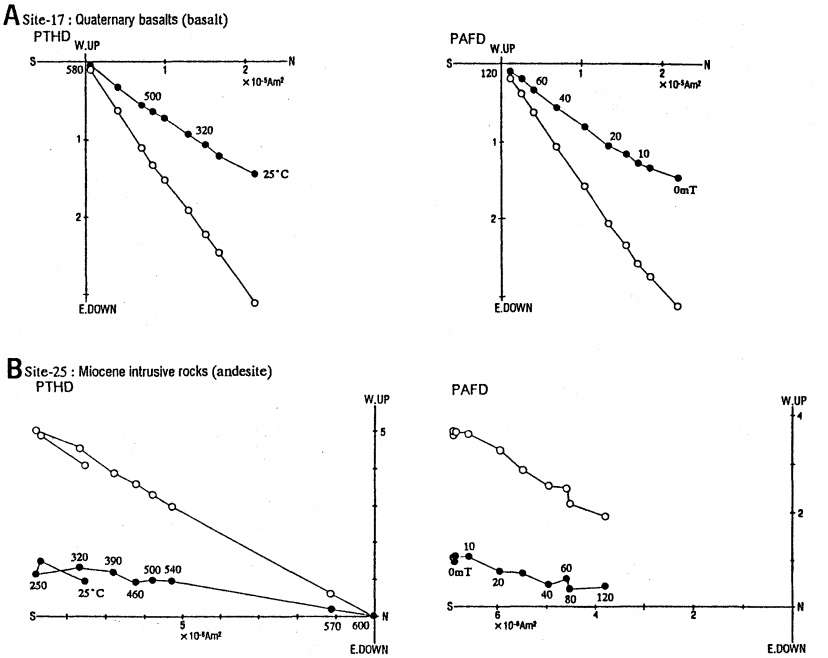


Fig. 4-1. Vector-demagnetization diagrams of pilot progressive demagnetization experiments by which one stable components were found. The demagnetization results are shown on the diagrams in in-situ coordinates. Solid and open symbols are projection on horizontal and N-S vertical planes, respectively. Unit of coordinates is bulk remanent intensity. Numbers attached to symbols are demagnetization levels. PTHD: progressive thermal demagnetization. PAFD: progressive alternating field demagnetization.

nents were isolated (Figs. 4-2 and 4-3). Sedimentary rock specimens from only five sites (sites 15, 50, 52, 57, and 68) and many igneous rock specimens yielded stable components decaying toward the origin in the higher temperature range up to 590°C or 610°C after the lower-temperature components were erased out (Fig. 4-2B, C, and D). Among the eight sites which provided the moderate-temperature components, we could obtain higher-temperature components decaying toward the origin from three sites (sites 24, 4 and 5; Fig. 4-3B and C).

Although the lower-temperature components were often observed during the AF demagnetization experiments, the AF treatment was hardly effective to isolate the moderate- and the higher-temperature components (Figs. 4-2 and 4.3). Only the pilot specimens of site 35 (diorite) yielded two components by the AF treatment as well as the thermal one (Fig. 4-2D). The results from the AF treatments of site 35 agreed very well with the thermal.

We could not obtain any stable components from some pilot specimens by either thermal or AF treatments because they showed extremely erratic magnetic behaviors (Fig. 4-4A) or no straight line segments on the diagrams (Fig. 4-4B). The latter magnetic behaviors were often observed in the pilot demagnetization results from granite sites, and suggested the presence of two or more components with overlapping unblocking temperatures and/or coercivity spectra.

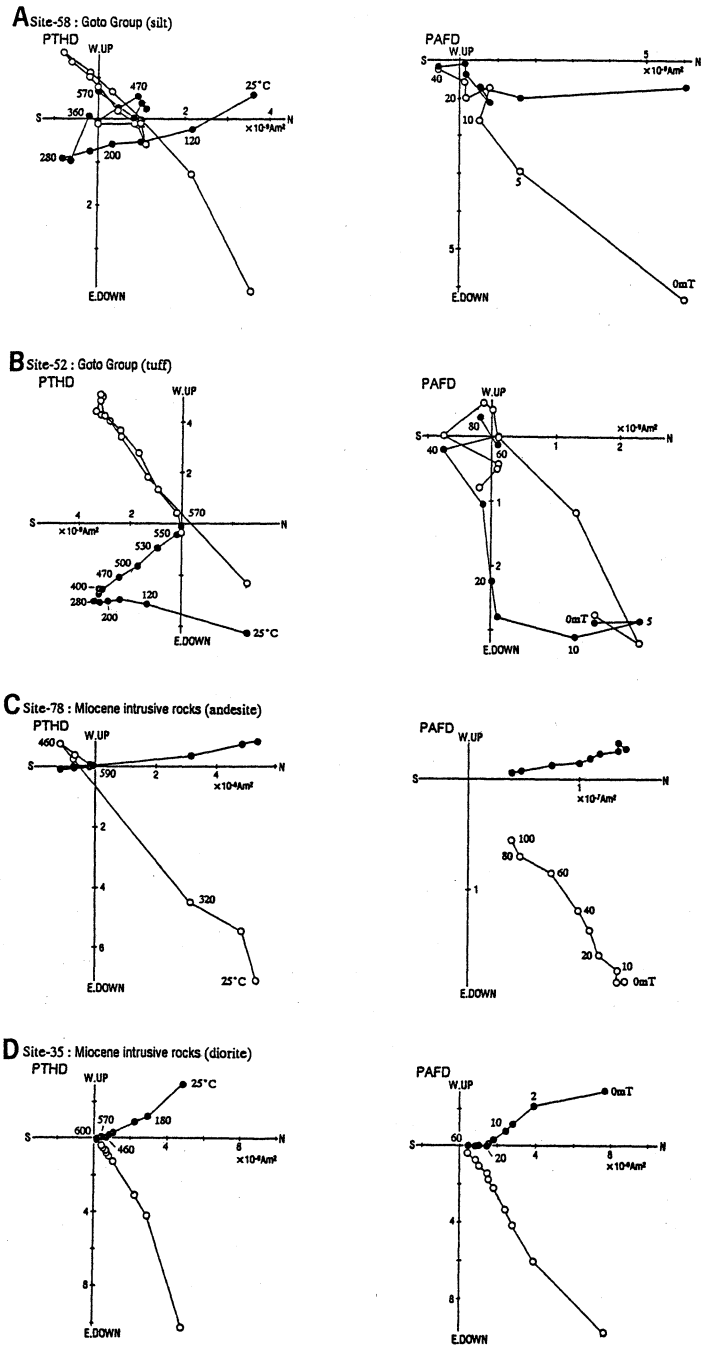


Fig. 4-2. Vector-demagnetization diagrams of pilot progressive demagnetization experiments by which the lower-temperature components were isolated. B, C and D are the examples that another components, the higher-temperature components, were also isolated by thermal demagnetization experiments. D is the example that two components were found by both thermal and AF demagnetization experiments.

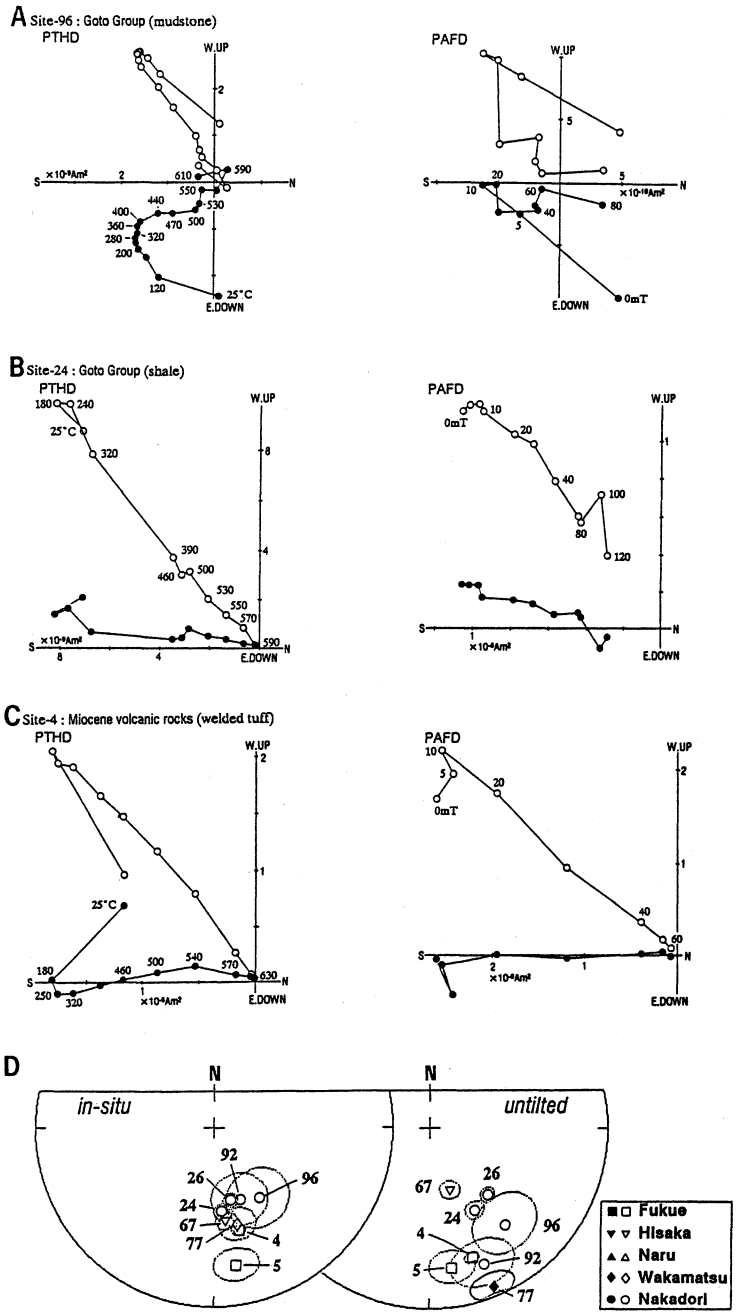


Fig. 4-3. Vector-demagnetization diagrams of pilot progressive demagnetization experiments by which the moderate-temperature components were isolated (A, B and C) and equal-area projections of site-mean directions of the moderate-temperature components before and after tilt correction (D). B and C are the examples that another components, the higher-temperature components, were also isolated by thermal demagnetization experiments.

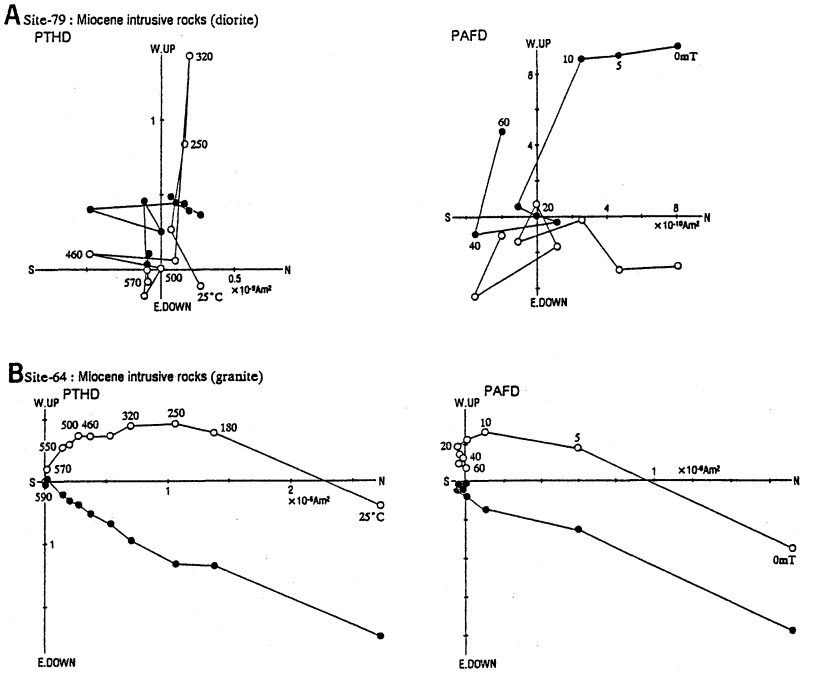


Fig. 4-4. Vector-demagnetization diagrams of pilot progressive demagnetization experiments by which no stable component was isolated.

In the following consideration, we focused on the remanent directions which were recognized as a linear trend toward the origin on the diagram at higher demagnetization steps. The stable component which did not decay toward the origin of the diagram was regarded as a secondary magnetic component. Pilot specimens from 37 sites yielded the stable components decaying toward the origin. All remaining specimens of the 37 sites were demagnetized at an optimum demagnetization step. Based on the pilot demagnetization results, the lowest demagnetization step was chosen as an optimum step among the steps at which the stable component was clearly recognized as a linear trend. We mostly used the thermal treatment for the optimum demagnetization. We used the AF treatment only when pilot demagnetization results from the thermal and AF treatments were in good agreement. Site-mean directions and associated statistical parameters of the stable magnetic components were calculated using the method of FISHER (1953). Among the 37 sites, sites 62 (rhyolite) and 88 (quartz porphyry) yielded the site-mean directions with larger α_{95} ($>30°$), while the other site-mean directions had smaller α_{95} values ($<20°$). We discarded the directions of sites 62 and 88 for further consideration. The remaining 35 site-mean directions are shown in Fig. 5 by equal-area projection and listed on Table 1.

3.1.2. Site-mean directions

Site-mean directions from the Quaternary basalts at five sites appeared to be close to the present geomagnetic field direction (Fig. 5(1)). A formation-mean direction was calculated from the five site-mean directions; $D=8.8°$, $I=54.2°$, $\alpha_{95}=11.0°$, and $k=49.6$.

We obtained 17 site-mean directions from various types of Miocene intrusive rocks

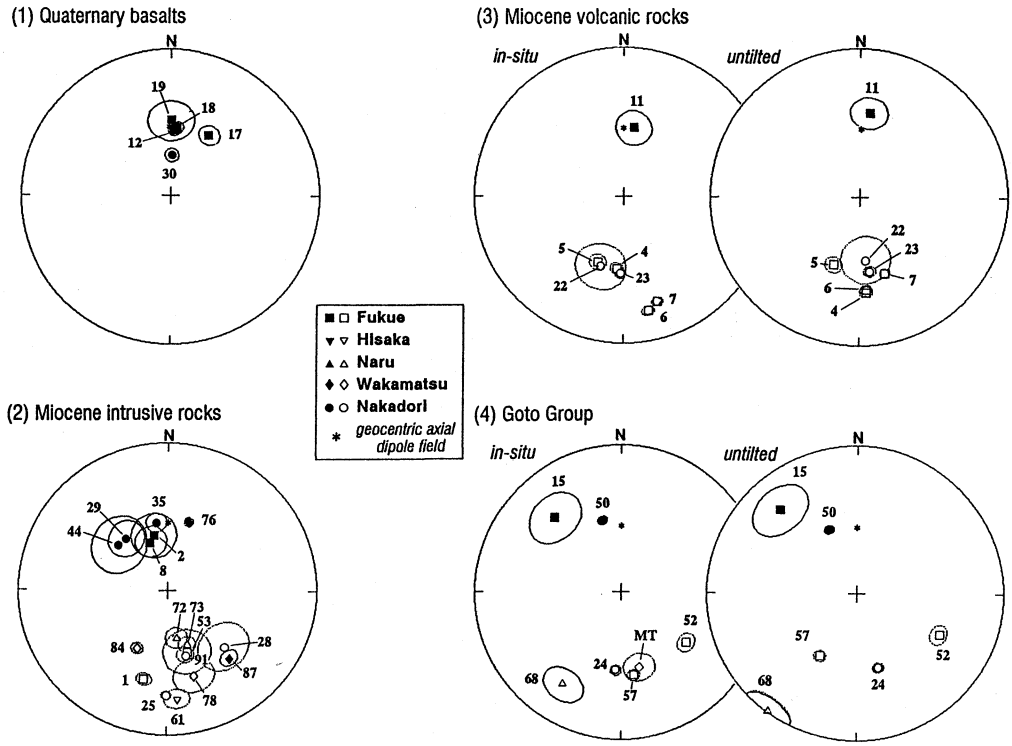


Fig. 5. Equal-area projections of site-mean directions. Ovals around the directions indicate 95% confidence limit. Numerals denote site numbers. Solid and open symbols are on the lower and upper hemispheres, respectively. (1): site-mean directions from Quaternary basalts, (2): site-mean directions from the Miocene intrusive rocks, (3): in-situ and untilted site-mean directions from the Miocene volcanic rocks, (4) in-situ and untilted site-mean directions from the Goto Group. MT is a mean of in-situ directions of the moderate-temperature components (D=165.8°, I=−46.0°, α_{95}=8.2°).

distributing on the whole Goto Islands (Fig. 5(2)). All site-mean directions except that of 87 (rhyolite) showed an antipodal relationship, approximately with the north-to-south trend. The site-mean direction of site 87 is anomalous, which may be attributed to a record of the transitional field during a geomagnetic reversal or a considerable displacement around site 87. We discarded the direction of site 87. A formation-mean direction was calculated from the remaining 16 site-mean directions; D=−12.6°, I=50.3°, α_{95}=8.2°, and k=21.4.

We obtained site-mean directions of the stable higher-temperature components from seven sites of welded tuff from the Miocene volcanic rocks (Fig. 5(3)). The five directions on the Fukue Island (sites 4, 5, 6, 7 and 11) showed a better antipodal relationship in a north-to-south trend after tilt corrections; a mean direction of the five directions had α_{95}=18.8° and k=17.5 before tilt correction, and α_{95}=11.3° and k=47.1 after tilt correction. We therefore regarded the higher-temperature components as the primary magnetization component acquired before the tilting of welded tuff beds, probably at the time of formation of the welded tuff beds. Two untilted site-mean directions on the Nakarodi Island (sites 22 and 23) were consistent with those on Fukue Island. An over-all

Table 1. Paleomagnetic results from the Goto Islands.

Island	Site	Rock type	in-situ		untilted		α_{95}	k	N	Demag. level	VGP	
			D	I	D	I					Lat	Lon
Quaternary basalts												
Fukue	12	basalt	2.7	53.0			3.1	476.4	6	460°C	87.5	−164.9
	17	basalt	32.5	49.4			5.7	96.3	8	30 mT	62.3	−145.0
	18	basalt	4.5	51.2			4.0	285.4	6	460°C	86.1	−130.5
	19	basalt	0.4	46.8			12.3	30.7	6	30 mT	85.4	−55.6
Nakadori	30	basalt	1.5	67.3			4.0	190.9	8	500°C	72.2	132.2
mean			8.8	54.2			11.0	49.6	5			
Miocene intrusive rocks												
Fukue	2	granite	−19.5	61.1			8.5	43.7	8	540°C	71.9	76.0
	8	granite	−14.4	58.3			12.4	16.1	10	20 mT	76.8	71.2
	53	granite	162.2	−52.9			13.1	89.2	3	460°C	75.1	46.9
	1	rhyolite	−165.8	−36.5			3.9	158.0	10	540°C	72.3	−100.6
Hisaka	61	granite	174.6	−25.4			6.4	142.1	5	20 mT	70.0	−35.6
Naru	72	granite	169.2	−62.4			6.0	101.6	7	530°C	76.2	94.2
	73	andesite	159.8	−56.9			4.7	379.7	4	530°C	72.9	60.3
Wakamatsu	76	quartz porphyry	17.3	50.0			2.5	601.2	7	30 mT	75.2	−137.5
	78	andesite	162.3	−37.9			10.5	53.8	5	460°C	70.5	6.9
	84	quartz porphyry	−153.2	−52.6			3.3	343.4	7	460°C	67.6	−149.1
	87	quartz porphyry	137.7	37.0			4.7	206.2	6	30 mT	23.0	−7.8
Nakadori	25	dacite	−179.9	−28.1			2.7	356.9	9	540°C	71.8	−51.2
	35	diorite	−10.1	51.5			5.4	107.2	8	30 mT	81.5	37.1
	29	granite	−39.1	52.7			9.9	32.3	8	500°C	57.5	50.5
	44	granite	−46.9	52.3			15.4	12.1	9	20 mT	51.0	52.2
	28	diorite	135.2	−44.2			13.8	13.2	10	500°C	49.8	41.3
	91	andesite	164.1	−51.4			4.6	172.4	7	30 mT	76.6	39.3
mean			−12.6	50.3			8.2	21.4	16			

Miocene volcanic rocks											
Fukue	4 welded tuff	−174.2	−48.8	175.6	−36.4	3.3	237.9	9	540°C	77.0	−32.6
	5 welded tuff	−158.9	−49.6	−158.9	−49.6	4.6	110.5	10	540°C	71.9	−140.0
	6 welded tuff	167.3	−20.8	175.6	−37.4	3.0	260.3	10	540°C	77.6	−31.7
	7 welded tuff	161.9	−25.2	161.1	−45.1	2.7	317.6	10	540°C	72.5	23.5
	11 welded tuff	8.6	51.9	6.2	42.2	9.8	28.5	9	500°C	80.0	−85.9
Nakadori	22 welded tuff	−177.7	−46.0	171.7	−48.3	3.2	226.0	10	500°C	81.9	12.9
	23 welded tuff	−162.3	−47.5	174.1	−54.8	13.5	15.4	9	500°C	84.7	65.0
mean		1.6	42.3			12.9	23.0	7			
				−2.3	45.4	8.2	55.1	7			
Goto Group											
Fukue	15 tuffaceous ss	−41.6	31.7	−42.3	23.8	14.2	130.0	5	390°C	46.4	21.2
	52 tuff	127.7	−42.8	116.2	−36.7	5.1	175.9	6	400°C	32.5	43.5
	57 sandstone	170.8	−42.2	−149.6	−49.2	3.3	215.2	10	470°C	64.0	−143.6
Naru	68 sandstone	−147.9	26.2	−143.5	−0.5	10.3	42.9	6	470°C	42.7	−105.1
Nakadori	24 shale	−176.0	−45.6	163.3	−46.4	3.0	331.4	8	500°C	74.6	22.9
	50 sandstone	−14.9	47.8	−23.6	50.3	2.7	330.2	10	500°C	69.9	40.4

Note: in-situ, untilted: site-mean (formation-mean) directions before and after tilt correction, respectively. D, I: declination and inclination in degrees, respectively. α_{95}: radius of 95% confidence circle in degrees. k: precision parameter. N: number of samples (sites) used to determine site-mean (formation-mean) direction, respectively. Demag. level: demagnetization level at which an optimum demagnetization was done. Lat, Lon: latitude (°N) and longitude (°E) of the virtual geomagnetic pole (VGP) position (north-seeking pole), respectively.

mean was calculated from the seven tilt-corrected directions; $D=-2.3°$, $I=45.4°$, $\alpha_{95}=8.2°$, and $k=55.1$. We regarded the mean direction as a paleomagnetic direction of the Miocene volcanic rocks on the Goto Islands.

We obtained the site-mean directions of higher-temperature components from six sites of the Goto Group; sites 15, 52 and 57 on the Fukue Island, sites 24 and 50 on the Nakadori Island, and site 68 on the Naru Island (Fig. 5(4)).

Based on the pilot demagnetization results (Figs. 4-2 and 4-3), the effect of secondary magnetization due to the recent geomagnetic field was completely erased out from the magnetizations of the higher-temperature components. The in-situ directions of four sites except for sites 24 and 57 were different from those of the moderate-temperature components (Fig. 5(4)). We can safely say that the higher-temperature components of the four sites could not be the secondary components which were found through the demagnetization experiments.

Among three site-mean directions on the Fukue Island (sites 15, 52 and 57), the untilted directions of sites 15 and 52 were approximately antiparallel, while the untilted direction of site 57 was different from those of other two sites. The in-situ direction of site 57 was concordant with the mean of the in-situ directions of moderate-temperature components (Fig. 5(4)). The possibility of complete remagnetization still remained on the magnetization of the higher-temperature component from site 57. The in-situ direction of site 24 on the Nakadori Island appeared to be close to the mean of the moderate-temperature components. However, we believe that the higher-temperature component from site 24 was free from the effect of the moderate-temperature component because we isolated both the moderate- and the higher-temperature components from the NRMs of pilot specimens of site 24 (Fig. 4-3B), and because the untilted directions of sites 24 and 50 on the Nakadori Island were antiparallel.

The untilted directions of sites 15, 52, 24, and 50 approximately showed an antipodal relationship with CCW deflection. The antipodal relationship implied that these directions represent the direction of the ancient geomagnetic field at the formation of the Goto Group. On the other hand, the untilted direction of site 68 on the Naru Island had a SW declination with extremely shallow inclination. The in-situ direction of site 68 was different from the directions of the secondary components found through demagnetization experiments in this study. If the direction of site 68 is the direction of the primary magnetic component, its untilted direction may be rather attributed to a record of the transitional field during a geomagnetic reversal.

3.1.3. Interpretation

Formation-mean directions of the Miocene intrusive rocks and Quaternary basalts were concordant with the paleomagnetic direction of the Miocene volcanic rocks. These three directions are approximately consistent with the expected direction of the geocentric axial dipole field at the Goto Island. It is suggested that the Goto Islands have been subjected to no significant tectonic displacement since the activity of the Miocene volcanic rocks (Fig. 6).

The untilted site-mean directions from four sites of the Goto Group on the Fukue and the Nakadori Islands, regarded as the directions of primary magnetization components, show a CCW deflection to the expected direction of the geocentric axial dipole field (Fig. 6). It is implied that tectonic displacement with rotation in a CCW sense occurred in the Goto Islands sometime after the formation of the Goto Group. The apparent discrepancy between the directions from Fukue Island (sites 15 and 52) and

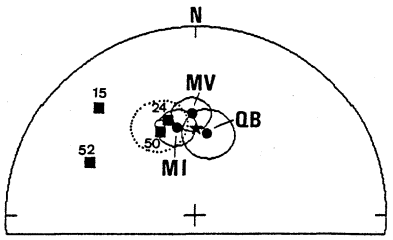


Fig. 6. Equal-area projection of formation-mean directions. Ovals around the directions indicate 95% confidence limit. QB: Quaternary basalts. MI: Miocene intrusive rocks. MV: Miocene volcanic rocks. Star symbol is the expected direction of the present geomagnetic field at the Goto Island. Square symbols with numerals are the untilted site-mean directions of four sites from the Goto Group (sites 15, 24, 50, 52). A open circle with oval indicates a mean paleomagnetic direction obtained from the Tsushima Islands (ISHIKAWA *et al.*, 1989).

Nakadori Island (sites 24 and 50) may be attributed to the difference of the CCW rotation angle between the two islands, or may be simply due to the small number of paleomagnetic data from the Goto Group.

3.2 Fission-track dating

3.2.1. Experiments

In separating zircon crystals from the samples of 22 sites, the samples generally yielded zircon crystals suitable for FT dating, although no suitable zircons were found in one rhyolite (site 1), three welded tuffs (sites 11, 22 and 23), and two diorites (sites 28 and 35).

FT ages of zircons from the remaining 16 sites were determined with the external detector method (EDM). The FT dating procedure using EDM in this study followed the procedure described in detail by TAGAMI *et al.* (1988). Spontaneous tracks in zircons were etched in the KOH–NaOH eutectic etchant (GLEADOW *et al.*, 1976) at 225±1° for 24–60 hours. Thermal neutron irradiation was performed for 15–30 minutes at the Irradiation Pit facility in the TRIGA II Reactor of Musashi Institute of Technology, at which facility the Cd ratio for Au is determined as 14±1 (T. Honda, personal communication).

FT ages were calculated using the zeta calibration approach (HURFORD and GREEN, 1983), in which an age calibration factor, zeta, is determined for a dosimeter glass using a set of well-established age standards. The zeta value used in this study was 342.1±6.2 (2σ) for NBS-SRM612, as reported by TAGAMI (1987). Errors were calculated using the "conventional analysis" given in GREEN (1981).

3.2.2. Results

Analytical data of zircon FT ages are presented in Table 2. A total of 20 measurements were carried out on 16 zircon samples. The statistical error of each determination did not contain an error component from the zeta constant, in order to compare the FT ages with each other. Repeated measurements on four samples (sites 2, 9, 36 on the Goto Islands, and site 39 on the Tsushima Islands) showed reasonable consistency with in 2σ errors.

Table 3 presents determined FT ages and rock types of the samples. A total of 16 zircon ages were determined on; four granite and two welded tuff samples in the Fukue

Table 2. Analytical data of FT age determination on zircons from the Goto and the Tsushima Islands.

Site	n	ρ_s	N_s	ρ_i	N_i	ρ_D	N_D	$T(\pm 1\sigma)$	$P(\chi^2)$
Goto Islands									
2-1	8	1.39	258	2.28	424	1.765	2954	18.3±1.5	90
2-2	7	1.63	225	3.15	436	1.757	2286	15.5±1.3	10
						weighted mean		16.9±1.0	
3	4	3.20	212	7.02	465	1.757	2286	13.7±1.2	50
4	3	1.79	104	4.29	249	1.757	2286	12.5±1.5	3
7	3	1.63	108	2.46	163	1.757	2286	19.9±2.5	30
8	6	3.43	431	6.26	786	1.765	2954	16.5±1.0	20
9-1	6	1.15	213	3.18	592	1.765	2954	10.9±0.9	6
9-2	10	1.81	496	3.78	1036	1.757	2286	14.4±0.8	4
						weighted mean		13.2±0.6	
25	4	6.02	406	11.44	771	1.757	2286	15.8±1.0	30
36-1	4	1.96	148	3.35	253	1.765	2594	17.6±1.9	10
36-2	6	2.06	285	3.83	529	1.757	2286	16.2±1.2	80
						weighted mean		16.7±1.0	
41	4	2.12	290	2.08	285	0.840	2342	14.6±1.3	75
42	8	1.08	304	2.01	565	1.765	2954	16.2±1.2	99
43	6	3.30	605	3.50	642	0.840	2342	13.5±0.8	20
44	11	0.96	401	1.95	812	1.765	2954	14.9±1.0	20
Tsushima Islands									
39-1	5	3.40	435	3.83	489	0.840	2342	12.8±0.9	30
39-2	7	3.85	636	7.35	1212	1.757	2286	15.7±0.8	2
						weighted mean		14.6±0.6	
43	10	2.08	796	3.77	1444	1.757	2286	16.5±0.8	25
47	5	4.25	750	4.76	840	0.840	2342	12.8±0.7	30
48	8	4.38	1547	4.17	1472	0.840	2342	15.1±0.6	3

Note: n: number of grains or fields counted. ρ_s: spontaneous track density of a sample ($\times 10^6$ cm^{-2}). N_s: counted track number for determining ρ_s. ρ_i: induced track density of a sample measured in muscovite detector ($\times 10^6$ cm^{-2}). N_i: counted track number for determining ρ_i. ρ_D: induced track density of SRM612 dosimeter measured in muscovite detector ($\times 10^5$ cm^{-2}). N_D: counted track number for determining ρ_D. T: FT age (Ma). $P(\chi^2)$: probability of obtaining the observed value of GALBRAITH's (1981) χ^2 parameter, for n degrees of freedom, where n=(number of crystals)−1, quoted to the nearest 5% or 10% except for values <10% or >95%.

Island, four granite, one welded tuff and one andesite samples in the Nakadori Island, and four granite samples in the Tsushima Islands. Although two determinations (sites 4 and 7) showed larger statistical errors, all of the ages were concordant within 2σ, averaging 15 Ma (Table 3).

3.2.3. Interpretation

All zircon FT ages from welded tuffs, an andesite, and granites on the Goto Islands were concordant, averaging about 15 Ma. The welded tuffs and andesite, which belong to lower rock units than granites in stratigraphic sequence (KAWAHARA *et al.*, 1984; MATSUI and KAWADA, 1986; Fig. 3), have little possibility of thermal effect by the granite intrusion because our microscope observation of their thin sections found no thermal effect and because those rocks providing concordant ages are distributed at

Table 3. Zircon FT ages determined and rock types of samples.

	Site	Rock type	FT age (Ma)($\pm 2\sigma$)
Goto Islands			
Miocene intrusive rocks			
Fukue	2	granite	16.9±2.0
	3	granite	13.7±2.3
	8	granite	16.5±2.1
	9	granite	13.2±1.3
Nakadori	41	granite	14.6±2.5
	42	granite	16.2±2.4
	43	granite	13.5±1.6
	44	granite	14.9±1.9
	25	andesite	15.8±2.0
Miocene volcanic rocks			
Fukue	4	welded tuff	12.5±3.0
	7	welded tuff	19.9±5.0
Nakadori	36	welded tuff	16.7±2.1
Tsushima Islands			
	39	granite	14.6±1.3
	43	granite	16.5±1.6
	47	granite	12.8±1.4
	48	granite	15.1±1.3

various distance from the granite stocks. The concordance of all the FT ages from the Goto Islands suggests, therefore, that the igneous activities of the Miocene volcanic rocks and the Miocene intrusive rocks on the islands occurred at about 15 Ma.

On the Tsushima Islands, one K–Ar and three zircon FT ages have been previously reported on the post-folding igneous rocks, showing Middle Miocene ages (KAWANO and UEDA, 1966; TAKAHASHI and HAYASHI, 1985, 1987). Precision and accuracy from these age data are, however, difficult to evaluate because the K–Ar dating was carried out over twenty years ago, and because the FT dating system was not calibrated by a set of age standards, which procedure is inescapable to a reliable FT dating (HURFORD and GREEN, 1982; MILLER *et al.*, 1985). Those previously reported data would give only a rough estimation for the timing of igneous activity on the Tsushima Islands. In contrast, zircon FT ages in the present study were determined by using a well-established calibration system. The four reliable FT ages of the granites, averaging about 15 Ma, suggest the post-folding igneous activity at about 15 Ma on the Tsushima Islands.

4. Discussion

4.1 CCW rotations of the Goto and Tsushima Islands and compressive deformation in the Tsushima Strait area

Paleomagnetic and geochronologic evidence from the Goto Islands suggests that the islands were subjected to a rotational movement in a CCW sense after the formation of the early to middle Miocene Goto Group and before the Miocene igneous activity at about 15 Ma on the islands. Our geochronologic data from the Tsushima Islands indicated that a CCW rotation of the islands of about 28° (ISHIKAWA *et al.*, 1989)

occurred at sometime after about 15 Ma. The timing of each CCW rotation is not determined exactly because the age control of the Goto Group is still poor and because there is no geologic sequence overlying Miocene igneous rocks on the Tsushima Islands. However, the CCW rotations of the two archipelagoes in the Tsushima Strait area imply that the Tsushima Strait area did not belong to Southwest Japan in terms of CW rotation at about 15 Ma, constraining the western margin of the CW-rotated block. A different tectonic regime should be suggested for the Tsushima Strait area in the formation process of the Japan Sea.

Pre-middle Miocene sediments on the Tsushima and Goto Islands, namely the Taishu and the Goto Group, are deformed by folding and faulting. The deformation features of the sediments are almost similar, showing N-S to NE-SW structural trends dominantly (UEDA, 1961; TESHIMA and YAMAMOTO, 1972; SHIMADA, 1977; KAWAHARA *et al.*, 1984). Seismic reflection profiles in the Tsushima Strait area show the existence of similar deformation features in the sea area (KATSURA and NAGANO, 1976; SAKURAI and NAGANO, 1976; NAGANO *et al.*, 1976); the sediments in the area are acoustically divided into five stratigraphic layers, and the lowest layer is correlated to the Taishu Group or Goto Group on the profiles. The deformation features are recognized in the lowest layer between the two archipelagoes. SHIMADA (1977) suggested that fold and fault systems of the Taishu Group was formed due to a horizontal compressive force in NW-SE direction. On the basis of the similarity of deformation features of the pre-middle Miocene sediments, it is suggested that the Tsushima Strait area was subjected to compressive deformation after the deposition of pre-middle Miocene sediments. Since Miocene igneous rocks unconformably overlie and/or intrude the deformation features of the sediments on the two archipelagoes, the FT ages of the igneous rocks determined in this study constrain the timing of the compressive deformation to be at sometime in the early to middle Miocene before about 15 Ma.

Geological data from exploratory wells and seismic reflection profiles on the southern margin of the Tsushima Basin in the northeast of the Tsushima Islands indicate that this region was subjected to a rapid subsidence under tension in the early to middle Miocene and subsequently an intensive deformation by folding and faulting under compression in the late Miocene (MINAMI, 1979; CHOUGH and BARG, 1987). The fold and fault systems in this region show a NE-SW structural trend almost similar to those in the Tsushima Strait area. Hence, the compressive deformation event in the Tsushima Strait area has been considered coeval with that on the southern margin of the Tsushima Basin in the late Miocene (INOUE, 1982). However, our geochronologic studies reveal that the Tsushima Strait area was subjected to a compressive deformation at the early to middle Miocene before about 15 Ma. It is therefore suggested that the compressive deformation in the Tsushima Strait area preceded the late Miocene compressive deformation on the southern margin of the Tsushima Basin and was rather coeval with the rapid subsidence on the southern margin of the Tsushima Basin.

In the Tsushima Strait area, several large scale faults with NNE-SSW to NE-SW and NW-SE trends are found by seismic investigations (INOUE, 1975; TOMITA *et al.*, 1975; KATSURA and NAGANO, 1976; SAKURAI and NAGANO, 1976; NAGANO *et al.*, 1976). Seismic reflection profiles show that those faults were formed or reactivated after the deposition of pre-middle Miocene sediments, that is, associated with the early to middle Miocene compressive deformation in the area (e.g., KATSURA and NAGANO, 1976). Among those faults, a NNE-SSW trending large scale fault zone exists lying off the

western coasts of the Tsushima and Goto Islands, which is referred to as the Tsushima-Goto tectonic line (KIMURA *et al.*, 1975; NAGANO *et al.*, 1976). The fault zone had an effect on the development of deformation of the Taishu Group on the Tsushima Islands (TOMITA *et al.*, 1975; SHIMADA, 1977; INOUE, 1982); the sedimentary rocks on the western coast of the islands are more intensively deformed than those on the eastern coast, and fold axes are dragged along the western coast and change in trend from NE-SW to NNE-SSW. These geologic features imply a sinistral motion of the fault zone during and/or after the compressive deformation of the Taishu Group (INOUE, 1982; HOSHINO, 1985).

The Tsushima and the Goto Islands can be regarded as faulted blocks bounded by those faults (KATSURA and NAGANO, 1976). The sinistral motion of the Tsushima-Goto tectonic line is preferable for the CCW rotations of the two archipelagoes because a rotational displacement in a CCW sense can result from a plastic drag and/or a block rotation along the sinistral fault (MACDONALD, 1980). Hence, the CCW rotations of the two archipelagoes may have taken place in response to the movement of the fault system in the Tsushima Strait area, especially the sinistral motion of the Tsushima-Goto tectonic line.

The Goto Islands were rotated CCW before about 15 Ma coeval with the early to middle Miocene compressive deformation event in the Tsushima Strait area, while the CCW rotation of the Tsushima Islands occurred after about 15 Ma. We can not exactly determine whether or not the timing of the CCW rotation was different between the two archipelagoes.

The movement of the fault system in the Tsushima Strait area was active at the time of the early to middle Miocene compressive deformation in the area (e.g., NAGANO *et al.*, 1976). It is highly possible that the sinistral motion of the Tsushima-Goto tectonic line occurred associated with the compressive deformation in the area. The CCW rotations of the two archipelagoes may have taken place almost coevally in response to the fault movement associated with the compressive deformation in the area. In this case, it is assumed that the Miocene igneous activity on the Tsushima Islands occurred slightly prior to that on the Goto Islands. This assumption may be allowed because such discrepancy of the timing of igneous activity would be smaller than the errors of radiometric age determination.

If the CCW rotation of the Tsushima islands occurred after that of the Goto Islands, the CCW rotation of the Tsushima Islands may not have been related to the early to middle Miocene compressive deformation in the Tsushima Strait area. Seismic reflection profiles show that a part of the Tsushima-Goto tectonic line off the Tsushima Islands was also active at the time of the late Miocene compressive deformation on the southern margin of the Tsushima Basin (INOUE, 1982). The deformation features on the region suggests a NW-SE trending compressive force (MINAMI, 1979; INOUE, 1982). Since the compressive force appears to be oblique to the fault zone, a sinistral motion of the fault zone may have also occurred due to the compressive force. Hence, the CCW rotation of the Tsushima Islands may have taken place associated with the late Miocene compressive deformation on the southern margin of the Tsushima Basin.

4.2 Implications for the opening mode of the Japan Sea

The compressive deformation event in the Tsushima Strait area occurred coeval with the rapid subsidence on the southern margin of the Tsushima Basin in the early to middle

Miocene. The rapid subsidence in this region has been attributed to the opening of the Japan Sea accompanied with the CW rotation of Southwest Japan at about 15 Ma (CHOUGH and BARG, 1987). Therefore, it is suggested that the Tsushima Strait area was subjected to compressive deformation during the opening stage of the formation process of the Japan Sea. The compressive tectonic situation in the area implies the convergence between the Korean Peninsula and Kyushu Island. Because the northern part of Kyushu Island has been regarded as a part of the CW-rotated Southwest Japan block (SHIMAZAKI *et al.*, 1981; OTOFUJI and MATSUDA, 1987) and because the timing of the compressive deformation event in the area was close to that of the CW rotation of the Southwest Japan block, it is implied that the western margin of the Southwest Japan block moved to the Korean Peninsula during the CW rotation of the block.

The sense of the relative motion of the western margin of the Southwest Japan block to the Korean Peninsula depends on the position of the rotation pivot of the block. The rotation pivot has been located to the west of Kyushu Island at 34°N and 129°E (OTOFUJI and MATSUDA, 1983; Fig. 7). OTOFUJI and MATSUDA (1983) estimated possible positions of the pivot which enables the Southwest Japan block to be rotated clockwise without a collision with neighboring regions. They located the pivot at 34°N and 129°E on the assumption that the rotation pivot was located near the end of the moving block. The convergent tectonic situation in the Tsushima Strait area during the CW rotation of Southwest Japan was not taken into their consideration. According to the pivot suggested by OTOFUJI and MATSUDA (1983), the western margin of the Southwest Japan block moves southward during its CW rotation, and a divergent tectonic situation is likely to be inferred around the Tsushima Islands as shown simplified in Fig. 7(a). To satisfy both, the opening of the Japan Sea and the convergence in the Tsushima Strait related to the CW rotation of the Southwest Japan block, the rotation pivot should have been located to the east of the Tsushima Strait area. For example, if the Southwest Japan block was rotated at about the pivot shown in Fig. 7(b), its western margin would move to the Korean Peninsula. These movements would result in compressive tectonics in the Tsushima Strait area and a sinistral motion of the Tsushima-Goto tectonic line as shown Fig. 7(b).

The position of the pivot controls the paleoposition of the Southwest Japan block prior to its CW rotation. Comparing the paleoposition of the Southwest Japan block in Figs. 7(a) and 7(b), the Southwest Japan block in Fig. 7(b) is located more southward than that in Fig. 7(a), and the back-arc space in Fig. 7(b) is much wider.

When Southwest Japan is rotated back to the eastern margin of Eurasia about the pivot at 34°N and 129°E, the Japan Sea is almost closed (OTOFUJI and MATSUDA, 1983; HAYASHIDA and TORII, 1988). It has been thus suggested that the Japan Sea was opened at about 15 Ma (OTOFUJI *et al.*, 1985c). However, the paleoposition of the block in Fig. 7(b) implies the possibility of the drifting of the Southwest Japan block prior to the CW rotation at about 15 Ma, that is, the initial opening of the Japan Sea occurred before about 15 Ma. Early Miocene lake and marine sediments are distributed on the Japan Sea coast of Southwest Japan, these sediments have been considered to have been deposited during the initial stage of the formation process of the Japan Sea (HUZIOKA, 1972; KOIZUMI, 1988; KANO and YANAGISAWA, 1989). Furthermore, evidence obtained by legs 127 and 128 of Ocean Drilling Program in the Japan Sea indicate that the Yamato Basin was formed by rapid subsidence during the early Miocene probably before about 19 Ma (LEG 127 AND LEG 128 SHIPBOARD SCIENTIFIC PARTIES, 1990). These geological data

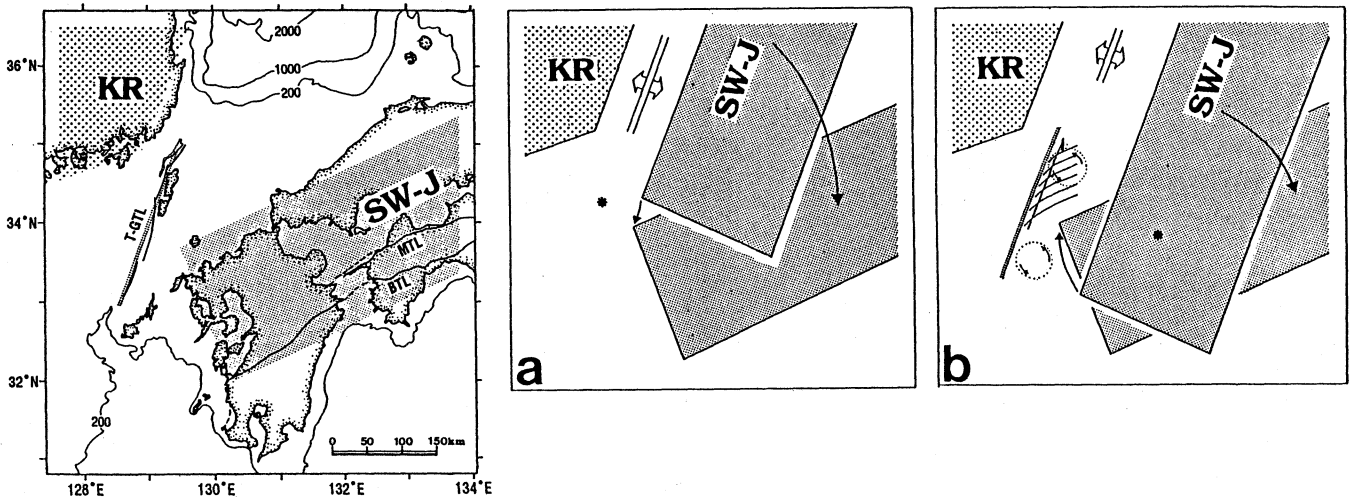


Fig. 7. Schematic diagram illustrating the manner in which divergent or convergent tectonics occurs in the Tsushima Strait area during CW rotation of the Southwest Japan block controlled by the position of the rotation pivot. Two shaded areas with different patterns represent the Korean Peninsula (KR) and the Southwest Japan block (SW-J), respectively. T-GTL: Tsushima-Goto tectonic line. MTL: Median Tectonic Line. BTL: Butsuzo Tectonic Line. (a) SW-J is rotated about a pivot at 34° N and 129° E. The western margin of SW-J moves away from KR. (b) A pivot is located to the east of the Tsushima Strait area so that the western margin of SW-J moves close to KR. The Tsushima Strait area will be subjected to a compressive tectonics during the CW rotation of SW-J; fold system will be formed, and a sinistral motion of T-GTL will take place.

support the paleoposition of the Southwest Japan block as shown in Fig. 7(b).

From the view point of compressive tectonics in the Tsushima Strait area, it is implied that during the formation process of the Japan Sea, the Southwest Japan block had been probably drifted away from the eastern margin of Eurasia before about 15 Ma in some manners which can not be identified by paleomagnetism, e.g., southward shifting with small or null CW rotation, and then the block was rotated clockwise at around 15 Ma about a pivot located to the east of the Tsushima Strait area.

5. Conclusions

1) Timing of Miocene igneous activity in the Goto and Tsushima Islands was determined about 15 Ma based on zircon FT ages from 16 out of 22 sites of Miocene igneous rocks in these islands.

2) Paleomagnetic directions of the Quaternary basalts, Miocene intrusive rocks, and Miocene volcanic rocks on the Goto Islands were determined as follows;

Quaternary basalts: $D=8.8°$, $I=54.2°$, $\alpha_{95}=11.0°$.
Miocene intrusive rocks: $D=-12.6°$, $I=50.3°$, $\alpha_{95}=8.2°$.
Miocene volcanic rocks: $D=-2.3°$, $I=45.4°$, $\alpha_{95}=8.2°$.

Untilted site-mean directions of primary magnetic components with high-blocking temperature were determined on four sites of sedimentary rocks of the Early to middle Miocene Goto Group. The directions were different from the expected direction calculated from the geocentric axial dipole field at the Goto Islands, showing CCW deflection in its declination.

3) The Goto Islands were probably subjected to a rotational movement in CCW sense sometime after the formation of the Goto Group and before the Miocene igneous activity at about 15 Ma, and the islands have not suffered significant tectonic displacement since about 15 Ma. The timing of CCW rotation of the Tsushima Islands after the Miocene igneous activity on the islands suggested by ISHIKAWA *et al.* (1989) was later than about 15 Ma.

4) The Goto and Tsushima Islands in the Tsushima Strait area did not belong to the Southwest Japan block in terms of CW rotation at about 15 Ma, constraining the western margin of the CW-rotated block.

5) The compressive deformation of pre-middle Miocene sediments in the Tsushima Strait area took place at sometime in the early to middle Miocene before about 15 Ma.

6) The compressive deformation in the Tsushima Strait area was coeval with the opening of the Japan Sea, accompanied with the CW rotation of the Southwest Japan blocks at about 15 Ma. This implies the convergence between the Korean Peninsula and the western margin of the Southwest Japan block during the CW rotation of the block.

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